The Impact of Makeup on Automated Face Recognition and Age Estimation Models Across Everyday and Extreme Conditions

1st van der Velde
Faculty of Technology, Hogeschool van Amsterdam
Amsterdam, The Netherlands
frank.van.der.velde@hva.nl

***Abstract*—Self-checkout systems in supermarkets face persistent challenges with age verification for alcohol sales. Automated facial age estimation offers a potential solution but may be unreliable when appearance is altered by makeup. This study evaluates how everyday and extreme synthetic makeup styles affect prediction accuracy. A MobileNetV3-Small model, fine-tuned on UTKFace, estimates continuous ages via regression. Two paired before-and-after datasets are used: FFHQ-Makeup for natural cosmetics and LADN for stylized extreme transformations. Age shifts are measured through changes in predicted apparent age and mean absolute error (MAE). Results quantify the sensitivity of age-estimation models to cosmetic variation and highlight implications for fairness and reliability in AI-based retail compliance.**

***Index Terms*—Age Estimation, Makeup Bias, Computer Vision, Facial Analysis, Alcohol Compliance, Apparent Age**

I. Introduction

Self-checkout systems are increasingly common in Dutch supermarkets, with Albert Heijn leading adoption to improve efficiency and reduce waiting times [1]. Alcohol sales remain problematic: between July 2023 and June 2024, the NVWA reported frequent failures in age verification [2]. Because fines are high, supermarkets promote responsible sales through policies like NIX18 [3], yet checks often remain incomplete in both physical and online contexts [4].

Manual verification at self-checkouts is time-consuming and error-prone, motivating interest in automated age estimation. Such systems could support compliance but face privacy and ethical barriers [1]. As technology matures, their adoption becomes more plausible, raising concerns about accuracy and fairness.

Makeup is an overlooked factor in this domain. By altering texture and contrast, cosmetics affect perceived age and can mislead computer-vision models [7]. While prior biometric studies examined these effects, their relevance for retail age verification remains unexplored. Fairness requires that people wearing makeup are not systematically misclassified, and reliability demands that underage users cannot exploit cosmetics to appear older.

This study therefore asks: *"What is the impact of makeup on automated face recognition and age estimation models across everyday and extreme conditions?"* A MobileNetV3-based age estimator is evaluated on paired synthetic before-and-after datasets representing natural and stylized makeup to assess how appearance changes affect prediction stability and compliance accuracy in AI-assisted alcohol sales.

II. Literature Review

A. Effects of Makeup on Visual Cues

Makeup alters key facial cues used for age perception, including skin tone evenness, local contrast, and texture visibility (e.g., wrinkles or pores) [5], [7]. These changes bias both human and algorithmic judgment by concealing aging signals or emphasizing youthful traits. Everyday cosmetics typically cause mild rejuvenation, while artistic or extreme styles (e.g., drag, horror) can produce stronger distortions that challenge automated systems. Studies show that makeup can lead to age overestimation in younger faces and underestimation in older ones by reshaping or masking critical cues [6].

B. Automated Facial Analysis and Model Vulnerabilities

Facial analysis models such as DEX and InsightFace achieve strong benchmarks yet remain sensitive to domain shifts from lighting, occlusion, and altered appearance [9]. Few works systematically test their robustness under varying makeup intensities or extreme stylization [5]. Cosmetics can act as adversarial alterations, reducing recognition or estimation accuracy known as "makeup attacks" [5] highlighting the need to assess visual domain robustness in applied contexts such as retail compliance.

C. Deep Learning for Age Estimation

Modern age estimation models use deep CNNs optimized for accuracy and computational efficiency. MobileNetV3 offers an effective balance between inference speed and regression stability, making it well suited for real-time retail applications [12]. When fine-tuned on datasets like UTKFace, such lightweight networks achieve competitive accuracy [16], though most prior studies test them only under natural conditions, without examining robustness to appearance shifts from cosmetics or synthetic makeup domains.

D. Dataset Limitations and Gaps

Common datasets like UTKFace and IMDB-WIKI provide demographic breadth but little cosmetic variability, limiting generalization. While some datasets target beautification or expression transfer, few analyze how makeup especially synthetic or stylized forms affects perceived or predicted age [7]. Controlled paired datasets such as FFHQ-Makeup and synthetic blend sets enable before-and-after comparisons under consistent conditions, supporting systematic analysis of these effects.

E. Reliability and Validity in Prior Studies

Reliable facial analysis requires controlled imagery and verified labels, yet many datasets remain unbalanced across gender, ethnicity, and cosmetic use, restricting fairness assessments [8]. Few works explicitly pair pre- and post-makeup samples, hindering precise measurement of cosmetic effects. Addressing these gaps is essential to evaluate algorithmic bias and performance consistency.

F. Research Gap and Relevance

Few studies explore extreme makeup effects within practical settings such as automated alcohol age verification. Misclassifications in these systems have legal and ethical consequences, especially under the Dutch Alcoholwet, which mandates ID checks for anyone appearing under 25. This study closes that gap by testing model robustness across natural, synthetic, and extreme makeup conditions, contributing to understanding of fairness, inclusivity, and reliability in AI-assisted retail compliance.

III. Methodology

A. Research Design

This research uses a within-subject design, comparing predicted ages of the same face before and after synthetic makeup application. Because the datasets are artificially generated and preserve identity, differences directly represent the model's response to makeup rather than to confounding factors such as lighting or pose.

Two datasets are analyzed:

- **FFHQ-Makeup**, representing natural everyday cosmetics.
- **LADN**, featuring high-intensity artistic makeup transfers created through a GAN-based style model.

Together, these datasets provide both realistic and extreme visual conditions under consistent identity control.

Fig. 1. Examples from FFHQ-Makeup showing bare-face and natural makeup variants.

Fig. 2. LADN examples of one person with four extreme makeup styles applied.

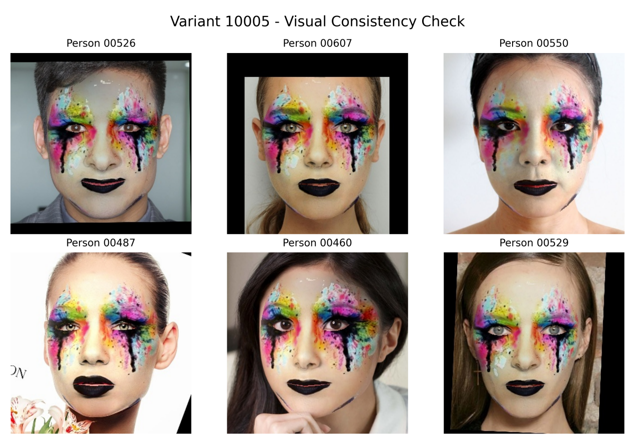

Fig. 3. The same LADN makeup variant (#10005) applied to six individuals, showing consistent transfer quality across identities.

These figures illustrate the visual consistency between bare and transformed versions and demonstrate how the LADN makeup transfer generalizes across different subjects.

B. Dataset Balancing and Sampling Strategy

To allow a controlled comparison, both datasets were balanced to contain approximately the same number of images (**37,690**). The LADN set includes 100 subjects, each with one bare-face image and 375 synthetic variants, while FFHQ-Makeup includes 6,281 subjects with one bare-face and five natural makeup versions each. This 1:5 ratio was maintained across both sets to equalize representation and control for sampling bias. Subjects were randomly selected to prevent overfitting to specific facial characteristics.

Dataset	People	Total Images	Avg Images/Person	Composition
FFHQ-Makeup	6,281	37,690	6.0	1 bare + 5 variants
LADN	100	37,690	376.9	1 bare + 375.9 variants

TABLE I
BALANCED DATASET STATISTICS. BOTH DATASETS CONTRIBUTE AN EQUAL TOTAL OF 37,690 IMAGES, ENSURING PARITY FOR COMPARATIVE EVALUATION.

This balancing strategy ensures that each dataset contributes equally to the evaluation, supporting a fair comparison of natural versus synthetic makeup effects on model predictions.

C. Model Choice

The model selection focused on achieving real-time efficiency and reliability for supermarket self-checkouts, where latency directly disrupts customer flow. Under these constraints, computational efficiency was prioritized over marginal benchmark gains. Large-scale models such as DEX [9], which rely on deep VGG-19 ensembles, were excluded due to their heavy memory and processing demands, making them impractical for low-power retail environments [10]. Instead, this research employs the lightweight MobileNetV3-Small architecture as the backbone for age estimation, aligning with the goal of testing makeup effects under realistic on-device conditions.

1) Rationale for Choosing MobileNetV3-Small: MobileNetV3-Small [12] represents a new generation of efficient CNNs optimized through neural architecture search and manual refinement for embedded inference. It integrates depthwise separable convolutions, squeeze-and-excitation (SE) modules, and h-swish activations to maximize representational power while minimizing computational cost. With only 66 MFLOPs, it achieves near-instant CPU inference without dedicated accelerators. Benchmarking shows that MobileNetV3 consistently outperforms MobileNetV1/V2 and other compact networks such as ShuffleNetV2 in both accuracy and latency [12], [18]. Its ability to capture fine visual cues like wrinkles and skin texture makes it particularly suitable for age estimation tasks that rely on subtle facial details.

2) Alternative Architectures: Several alternatives were evaluated to ensure that the chosen model is both performant and defensible in light of existing literature and deployment goals. VGG-19 and ResNet-based models (e.g., DEX [9], InsightFace) deliver high accuracy but have tens of millions of parameters and excessive inference latency. EfficientNet-B0 [11] achieves strong results yet its compound scaling increases computational overhead on non-GPU devices. ShuffleNetV2 [13] excels in speed but sacrifices expressive capacity, limiting regression performance on fine-grained tasks. MobileNetV2 improves efficiency but lacks the SE modules and h-swish activations that make V3 more accurate at nearly identical cost [12].

Given these trade-offs, MobileNetV3-Small provides the best balance between speed, accuracy, and deployability, making it well suited for real-time age estimation in self-checkout systems.

D. Age Estimation and Training Procedure

The age estimation task was formulated as a regression problem predicting continuous apparent age values, avoiding the discretization errors of class-based systems and producing smoother, more realistic results [14]. The final layer of the MobileNetV3-Small model was replaced with a single linear neuron optimized using Mean Squared Error (MSE) loss, which penalizes large deviations and stabilizes convergence to ensure a consistent mapping between facial features and age.

Training was conducted using the UTKFace dataset [21], a widely adopted benchmark containing over 20,000 labeled faces aged 0–116 with diverse poses, expressions, and lighting conditions. Compared to larger datasets such as IMDB-WIKI [19] and MORPH II [20], UTKFace offers a more balanced age distribution and less stylized imagery, supporting stronger generalization to real-world scenarios. Although its age labels are automatically estimated, this limitation has limited impact since the study examines relative prediction shifts rather than absolute accuracy. Ethnicity labels were excluded due to inconsistent definitions to avoid introducing bias.

Transfer learning was applied in two stages. First, convolutional layers were frozen and only the regression head was trained. Next, upper convolutional blocks were unfrozen and fine-tuned with a lower learning rate to refine high-level representations. Early stopping, adaptive learning-rate scheduling, and light augmentations (horizontal flipping, brightness adjustment, and rotation) promoted stable convergence and robustness.

E. Preprocessing and Evaluation

All images were cropped, aligned, and resized to 224×224 pixels, normalized to the ImageNet mean and variance, and filtered to retain only single, frontal, unobstructed faces with visible eyes and neutral expressions. These preprocessing steps minimize confounding variation from pose, occlusion, or lighting and improve reliability when applying synthetic makeup transformations later in the analysis.

For each dataset, the following evaluation steps were applied:

1) Predict baseline (no-makeup) age for each identity.
2) Predict the corresponding synthetic makeup variants.
3) Compute the difference between predicted ages before and after makeup.

Where ground-truth labels were available, Mean Absolute Error (MAE) was reported. The mean age shift (ΔAge) was also computed to capture systematic changes across datasets.

F. Reliability and Validity

Reliability was ensured through identical preprocessing, fixed random seeds, and consistent model parameters. To prevent data leakage, all before-and-after image pairs of the same individual were placed in the same data partition, ensuring that identities never appeared across train and test sets. Validity was strengthened through a paired within-subject design using complementary datasets FFHQ-Makeup for naturalistic styles and LADN for extreme, stylized makeup allowing the analysis to isolate appearance-based variation. Although the datasets might have limited balance for demographics, the paired setup mitigates bias by keeping network parameters constant while varying only visual conditions. Together, these design choices enhance the internal validity and reproducibility of the findings while acknowledging external limitations.

IV. Results

A. Prediction Trends Across Datasets

Dataset	Mean Age Delta (yrs)	Std Dev	Median	Range (yrs)	Younger (%)	Older (%)
FFHQ	-6.08 ± 14.35	14.35	-2.39	[-72.95, 44.36]	41.8	14.5
LADN	+1.37 ± 5.36	5.36	+1.21	[-41.54, 52.16]	2.7	6.3

TABLE II
AGE ESTIMATION DIFFERENCES ACROSS DATASETS, USING THE MODEL MAE (5.21 YEARS) AS A REFERENCE FOR PERCEPTIBLE CHANGE.

The LADN dataset consists of synthetically generated before-and-after image pairs. It primarily contains younger faces between 10 and 40 years old, which limits the variation in age-related facial features. The mean change of +1.37 years suggests a slight aging tendency after synthetic makeup application. However, no very extreme changes were observed, even despite the limited diversity in ages. Since the paired images are identical except for makeup, even smaller shifts below the model's mean absolute error (MAE) of 5.21 years can still represent a real effect. In this dataset, the changes remain modest and well within expected model variation.

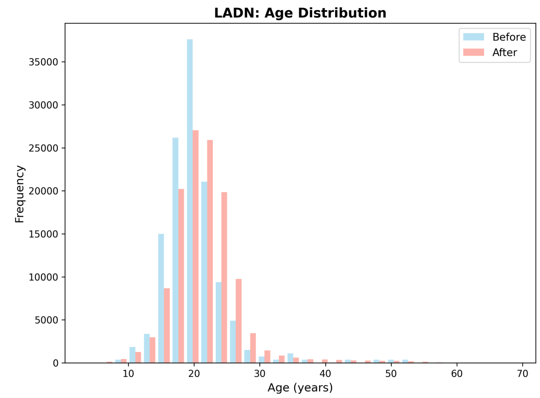

Fig. 4. Age distribution in the LADN dataset. The limited range (10–40 years) constrains the strength of conclusions.

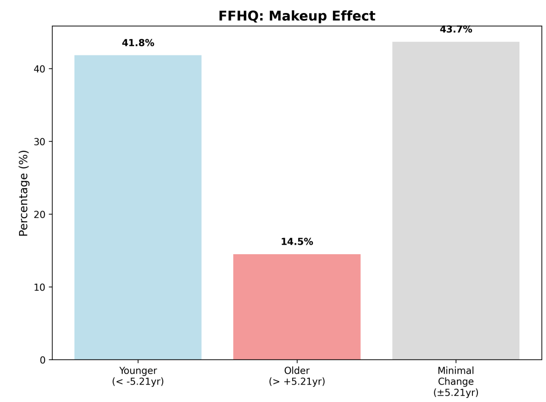

Fig. 5. Distribution of significant makeup effects in the FFHQ dataset.

B. Makeup-Induced Changes in FFHQ

The FFHQ dataset, which contains natural images with makeup applied synthetically, shows a clearer and more asymmetric pattern. A large portion of samples shift toward younger predicted ages, while a smaller fraction appears older. The overall rejuvenation trend supports the hypothesis that the model is sensitive to makeup-induced texture and contrast changes. Because the experiment uses paired before/after versions of the same face, these differences reflect the model's response to appearance alterations rather than identity or demographic factors.

C. Variance and Reliability of Predictions

When comparing all prediction differences to the model's baseline MAE, 76.5% remain within ±5.21 years. This indicates that the overall statistical reliability of the model is not substantially affected by makeup. Although the paired design amplifies sensitivity to small visual changes, the global variance remains limited. The effect is therefore measurable but not destructive to model stability.

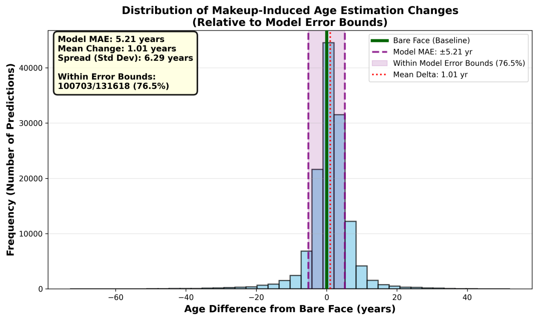

Fig. 6. Distribution of makeup-induced prediction shifts relative to the model's MAE (±5.21 years).

D. Relationship Between Makeup Effect and Initial Age

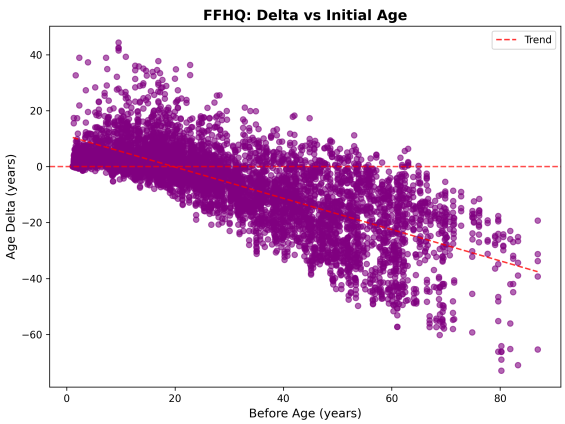

Fig. 7. Relationship between predicted baseline age and makeup-induced delta in FFHQ.

The relationship between initial predicted age and the resulting delta shows a clear gradient. Very young faces tend to receive slightly higher predictions after makeup, while older faces appear younger. This forms a continuous correlation from aging effects in youth to rejuvenation effects in older individuals. The result confirms that the model's response to makeup is not random but systematically related to the apparent age of the subject before transformation.

E. Threshold Crossing Effects (25-Year ID-Check Boundary)

For the supermarket use case, the 25-year threshold is relevant because Dutch regulations require ID verification when a person appears younger than that age. The threshold-crossing analysis shows that a notable number of FFHQ samples move across this boundary due to makeup: 235 individuals who initially appeared 25 years or older were predicted below 25 after transformation, while 86 moved upward. This implies that a significant group would unnecessarily trigger an age check, which could affect customer experience. Conversely, a smaller but still relevant number might bypass an ID check because of rejuvenating effects, highlighting the model's sensitivity near decision boundaries.

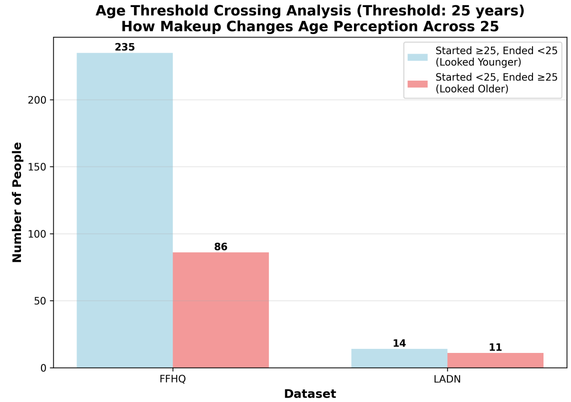

Fig. 8. Individuals crossing the 25-year ID-check threshold after makeup.

F. Summary of Findings

Across both datasets, the experiment using synthetic before-and-after makeup pairs shows that appearance changes can meaningfully influence age estimation. The LADN dataset exhibits mild effects without extreme deviations, while FFHQ demonstrates stronger variation, particularly among very young and older individuals. Although the statistical reliability of the model remains stable, the FFHQ threshold-crossing analysis reveals a practical implication: makeup can cause both unnecessary ID checks and occasional missed detections in automated retail systems. These findings emphasize the need for awareness of visual variability when applying computer vision models in regulated age-verification contexts.

G. Discussion Transition

The results highlight that age estimation under makeup is affected more by visual noise than by consistent bias, emphasizing the model's general stability. However, the observed nonlinear relationship with baseline age introduces subtle fairness concerns: older adults tend to appear younger, while very young individuals may appear slightly older after makeup. These tendencies, although within statistical error, may still influence automated systems in practice. The following discussion addresses these implications for fairness, bias mitigation, and system design in real-world age verification contexts.

V. Discussion

A. Interpretation

The results indicate that synthetic makeup alters age estimation in a consistent but direction-dependent way: younger faces appear slightly older, while older faces appear younger. Because paired before-and-after images were used, even sub-MAE shifts reveal meaningful visual effects, confirming that the model systematically responds to contrast and texture changes introduced by makeup.

B. Implications for Retail and Ethics

For retail age verification, these effects have ethical implications. The 25-year threshold analysis shows that cosmetics can cause unequal treatment, where individuals wearing makeup are either asked for identification more often or bypass checks they would otherwise receive. Such bias stems not from personal traits but from the model's sensitivity to visual alteration, influencing fairness and customer experience in systems supporting the *Alcoholwet*.

C. Reliability and Future Work

Most predictions stayed within normal model variance, but the study's synthetic scope and limited demographic diversity restrict generalization. Future work should test real before-and-after data to confirm these trends, replicate or extend the LADN setup with extreme styles, and include broader age and skin-tone representation. Complementary qualitative analysis could further reveal whether human perception aligns with model shifts, clarifying how cosmetic-induced changes affect fairness and reliability in applied computer-vision systems.

VI. Conclusion

This study demonstrates that makeup, even in everyday form, can meaningfully shift automated age estimation outcomes. Although most predictions remain within the model's expected error range, systematic trends appear: younger faces are estimated slightly older and older faces younger. These shifts can cause both unnecessary and missed ID checks around the 25-year boundary, which is relevant for retail compliance and customer experience.

Overall, the model remains statistically stable but perceptually sensitive to appearance changes. Further research using real data, broader age diversity, and cross-tone evaluation is needed to fully understand the impact of cosmetics on algorithmic fairness. A qualitative perspective would complement these findings and provide a clearer view of how makeup influences both machine and human perception of age.

References

[1] Avrotros. *"Mag dit zomaar?" Vragen over camera's bij de zelfscankassa.* 23 May 2025. [Online]. Available at: https://radar.avrotros.nl/artikel/mag-dit-zomaar-vragen-over-cameras-bij-de-zelfscankassa-61602

[2] Nederlandse Voedsel en Warenautoriteit (NVWA). *Inspectieresultaten Alcoholwet juli 2023 – juni 2024.* 2024. [Online]. Available at: https://www.nvwa.nl/onderwerpen/alcoholverkoop/inspectieresultaten/2024/alcoholwet-juli-2023-juni-2024

[3] Albert Heijn. *Duurzaamheidsverslag 2022 – NIX18.* 2022. [Online]. Available at: https://duurzaamheidsverslag.ah.nl/2022/nix18

[4] Nederlandse Voedsel en Warenautoriteit (NVWA Breuer) en Intraval. *Factsheet Online alcohol bestellen door jongeren 2024.* 2025.

[5] Kotwal, M., Zaveri, M., Shah, M. *Detection of Age-Induced Makeup Attacks on Face Recognition Systems Using Multi-Layer Deep Features.* IEEE, 2019.

[6] Gavas, E. B., Hegde, C., Memon, N., Banerjee, S. *DiffClean: Diffusion-based Makeup Removal for Accurate Age Estimation.* arXiv preprint, 2025.

[7] Chen, C., Dantcheva, A., Ross, A. *Impact of Facial Cosmetics on Automatic Gender and Age Estimation Algorithms.* In: *Proceedings of the International Conference on Computer Vision Theory and Applications (VISAPP)*, Lisbon, Portugal, 2014.

[8] Clapes, A., Bilici, O., Temirova, D., Avots, E., Anbarjafari, G., Escalera, S. *From Apparent to Real Age: Gender, Age, Ethnic, Makeup, and Expression Bias Analysis in Real Age Estimation.* In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, Salt Lake City, USA, 2018.

[9] Rothe, Rasmus, Guillaumin, Matthieu, Van Gool, Luc. *DEX: Deep Expectation of Apparent Age From a Single Image.* Available at: https://www.cv-foundation.org//openaccess/content_iccv_2015_workshops/w11/papers/Rothe_DEX_Deep_EXpectation_ICCV_2015_paper.pdf

[10] Howard, Andrew G., Zhu, Menglong, Chen, Bo, Kalenichenko, Dmitry, Wang, Weijun, Weyand, Tobias, Andreetto, Marco, & Adam, Hartwig. *MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications.* Available at: https://arxiv.org/abs/1704.04861.

[11] Tan, Mingxing, & Le, Quoc V. *EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks.* Available at https://proceedings.mlr.press/v97/tan19a.html?ref=ji

[12] Howard, Andrew, Sandler, Mark, Chen, Bo, Wang, Weijun, Chen, Liang-Chieh, Chen, Ming-Hsuan, Zhu, Yukun, Pang, Ruoming, Vasudevan, Vijay, Le, Quoc V., Adam, Hartwig, & Han, Xiaokai. *Searching for MobileNetV3.* Available at https://openaccess.thecvf.com/content_ICCV_2019/html/Howard_Searching_for_MobileNetV3_ICCV_2019_paper.html

[13] Ma, Ningning, Zhang, Xiangyu, Zheng, Hai-Miao, & Sun, Jian. *ShuffleNet V2: Practical Guidelines for Efficient CNN Architecture Design.* Available at https://openaccess.thecvf.com/content_ECCV_2018/html/Ningning_Light-weight_CNN_Architecture_ECCV_2018_paper.html

[14] Shikha Prasher; Leema Nelson; S. Gomathi *Automated Eye Disease Classification using MobileNetV3 and EfficientNetB0 Models using Transfer Learning* DOI: https://ieeexplore.ieee.org/abstract/document/10235193.

[15] Gil Levi, Tal Hassner *Age and gender estimation with deep convolutional neural networks.* Available at https://www.cv-foundation.org/openaccess/content_cvpr_workshops_2015/W08/html/Levi_Age_and_Gender_2015_CVPR_paper.html

[16] Kocoń M, Pawlukiewicz S *Age Estimation and Gender Classification from Facial Images.* Available at: https://www.mdpi.com/2076-3417/15/18/10212.

[17] Hoang, Van-Thanh and Jo, Kang-Hyun *Practical Analysis on Architecture of EfficientNet* Available at: https://ieeexplore.ieee.org/abstract/document/9538782.

[18] Zhang, Xiangyu and Zhou, Xinyu and Lin, Mengxiao and Sun, Jian *ShuffleNet: An Extremely Efficient Convolutional Neural Network for Mobile Devices* Available at: https://openaccess.thecvf.com/content_cvpr_2018/html/Zhang_ShuffleNet_An_Extremely_CVPR_2018_paper.html.

[19] Rothe, Rasmus, Timofte, Radu, Van Gool, Luc. *IMDB-WIKI – Dataset and Benchmark for Age and Gender Estimation.* Associated papers: Rothe, R., Timofte, R., Van Gool, L. *Deep Expectation of Real and Apparent Age from a Single Image Without Facial Landmarks.* International Journal of Computer Vision, 126(2–4), 144–157, Springer, 2018. Rothe, R., Timofte, R., Van Gool, L. *DEX: Deep EXpectation of Apparent Age from a Single Image.* In: *IEEE International Conference on Computer Vision Workshops (ICCVW)*, December 2015. Dataset available at: https://data.vision.ee.ethz.ch/cvl/rrothe/imdb-wiki/

[20] Bingham, G., Yip, B., Ferguson, M., Nansalo, C.; Mentors: Chen, C., Wang, Y., Kling, T. *MORPH-II: Inconsistencies and Cleaning Whitepaper.* NSF-REU Site at UNC Wilmington, Summer 2017. Available at: https://libres.uncg.edu/ir/uncw/f/wangy2017-1.pdf

[21] Zhang, Zhifei, Song, Yang, Qi, Hairong. *Age Progression/Regression by Conditional Adversarial Autoencoder.* In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, IEEE, 2017. Dataset available at: https://susanqq.github.io/UTKFace/